Exploratory Data Analysis on a Cars Dataset

Clean, explore, visualise, and explain meaningful patterns in automobile data

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Batch	Artificial Intelligence - October B - 2026 Batch
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Project type	Individual project - Student Project Assignment
Dataset source	Kaggle - "Car Features and MSRP" (file: data.csv)
Dataset size	11,914 rows x 16 columns (11,184 rows after cleaning)

2. Objective

The purpose of this project is to carry out an Exploratory Data Analysis (EDA) on a cars dataset: to understand its structure, improve its quality, find patterns and unusual records, and communicate the results through clear charts. No machine-learning model is used. The questions selected for this analysis are:

- Which car brands and transmission types are the most common?
- How are car prices distributed, and where are the outliers?
- Do automatic and manual cars differ in price?
- Which numerical features are strongly correlated with each other and with price?
- Which brands have the highest prices, and which records are unusual?

Two of the suggested questions were not used because the dataset cannot support them: this dataset has no mileage (odometer) column, and its price is the manufacturer's list price (MSRP), not a resale price.

3. Dataset Overview

The dataset (Kaggle, "Car Features and MSRP") contains **11,914 rows and 16 columns** describing cars from model years 1990 to 2017: 8 categorical (object) columns and 8 numerical columns (3 float, 5 integer).

Column	Type	Description
Make	object	Car brand (48 brands)
Model	object	Car model name (915 models)
Year	int	Model year (1990-2017)
Engine Fuel Type	object	Fuel the engine uses (10 types)
Engine HP	float	Engine horsepower
Engine Cylinders	float	Number of cylinders (0 for electric cars)
Transmission Type	object	Manual, automatic, automated manual, direct drive, unknown
Driven_Wheels	object	Front, rear, all or four wheel drive
Number of Doors	float	2, 3 or 4 doors
Market Category	object	Market labels such as Luxury, Performance, Crossover
Vehicle Size	object	Compact, Midsize or Large
Vehicle Style	object	Body style (16 styles, e.g. Sedan, 4dr SUV)
highway MPG	int	Fuel efficiency on highways
city mpg	int	Fuel efficiency in the city
Popularity	int	Popularity score of the brand/model
MSRP	int	Manufacturer's suggested retail price (USD)

Sample records (first 5 rows, selected columns)

Make	Model	Year	HP	Cyl.	Transmission	Size	Hwy MPG	MSRP
BMW	1 Series M	2011	335	6	MANUAL	Compact	26	46,135
BMW	1 Series	2011	300	6	MANUAL	Compact	28	40,650
BMW	1 Series	2011	300	6	MANUAL	Compact	28	36,350
BMW	1 Series	2011	230	6	MANUAL	Compact	28	29,450
BMW	1 Series	2011	230	6	MANUAL	Compact	28	34,500

Numerical columns: Year, Engine HP, Engine Cylinders, Number of Doors, highway MPG, city mpg, Popularity, MSRP. Categorical columns: Make, Model, Engine Fuel Type, Transmission Type, Driven_Wheels, Market Category, Vehicle Size, Vehicle Style.

4. Data Quality and Cleaning

The quality checks (*isnull().sum()*, *duplicated().sum()*, *unique()* and *value_counts()*) found the issues below. Each decision and its reason is listed.

Issue found	Count	Decision and reason
Duplicate rows	715	Removed with drop_duplicates(); identical rows add no new information.
Engine Fuel Type missing	3	Rows dropped; only 3 rows, too few to matter and no safe value to fill.
Engine HP missing	69	Filled with the column median; the median is not affected by very powerful cars.
Engine Cylinders missing	30	Filled with the column median.
Number of Doors missing	6	Filled with the mode (most common value, 4 doors) because doors is a category-like count.
Market Category missing	3,742	Filled with "Not Specified"; about 30% missing, so guessing values would be unreliable.
Transmission = UNKNOWN	12	Rows removed; the label gives no usable information.
Column names inconsistent	16	Renamed to lowercase with underscores (e.g. highway_mpg) for easy use.
Skewed price (MSRP)	-	Log transformation (np.log1p) applied for the log-scale box plot and the heatmap.
New feature	-	car_age = 2017 - year, because the data ends at model year 2017.
Outliers in MSRP	-	Kept. Very expensive cars (e.g. Bugatti Veyron 16.4, Lamborghini Reventon) are real luxury cars, not errors.

Dataset size before and after cleaning

	Before cleaning	After cleaning
Rows	11,914	11,184
Columns	16	17 (car_age added)
Duplicate rows	715	0
Missing values (total)	3,850	0

In total 730 rows (715 duplicates, 3 missing fuel type and 12 unknown transmission, counted after duplicates were handled in sequence) were removed.

5. Descriptive Analysis

Summary statistics of the cleaned numerical columns (MSRP shown in original dollars):

Column	Mean	Std	Min	Median	Max
year	2010.7	7.2	1990	2015	2017
engine_hp	253.3	109.8	55	237	1001
engine_cylinders	5.67	1.80	0	6	16
highway_mpg	26.6	9.0	12	25	354
city_mpg	19.7	9.2	7	18	137
popularity	1559.8	1446.1	2	1385	5657
msrp	41,973	61,562	2,000	30,705	2,065,902

Summary of the main categorical columns:

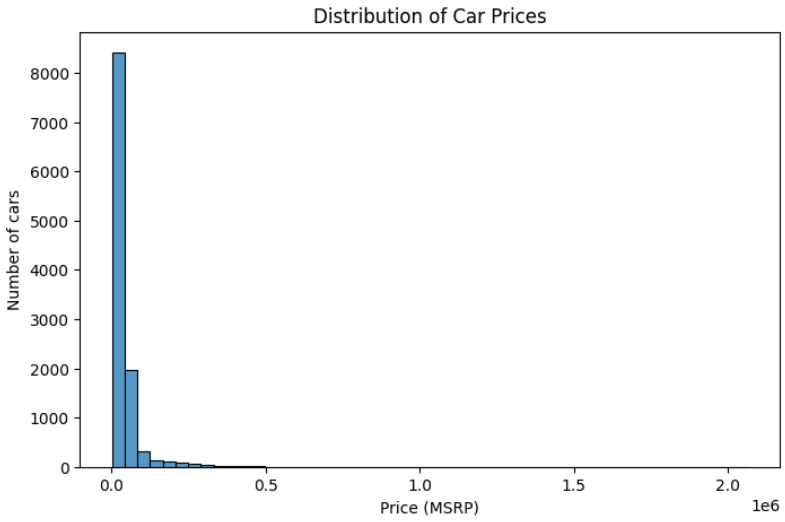
Column	Unique values	Most common value	Share of records
make	48	Chevrolet (1,083)	9.7%
transmission_type	4	AUTOMATIC (7,929)	70.9%
engine_fuel_type	10	regular unleaded (6,646)	59.4%
driven_wheels	4	front wheel drive (4,347)	38.9%
vehicle_size	3	Compact (4,389)	39.2%
vehicle_style	16	Sedan (2,839)	25.4%

Important observations: (1) The average price (41,973) is well above the median (30,705) and the standard deviation (61,562) is larger than the mean, which means a small number of very expensive cars pull the average up (skewness about 11.6). (2) The highway MPG maximum of 354 is far above the typical value of 25 and is an unusual record. (3) Engine cylinders has a minimum of 0, which belongs to electric cars (66 records) and is therefore valid.

6. Visual EDA

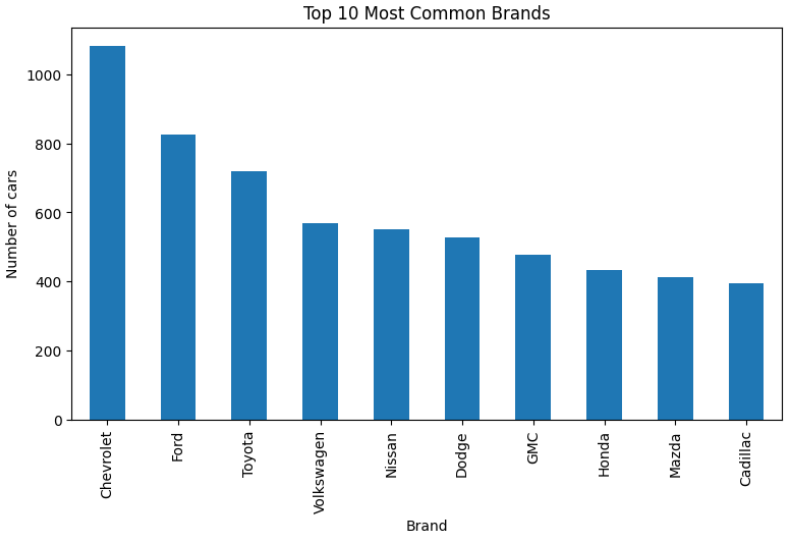
Six charts of five different types (histogram, bar chart, count plot, box plot, heatmap) are shown below, each with an observation.

Figure 1: Distribution of Car Prices (Histogram)



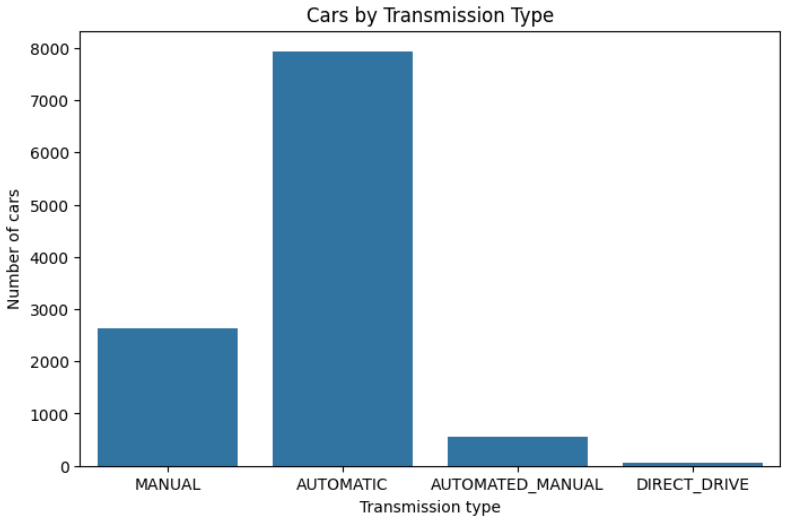
Observation: Prices are strongly right-skewed. About three quarters of all cars (roughly 8,400 of 11,184) fall in the first bar, from about 2,000 to 43,000, and the bars become very small after that. A few extremely expensive cars stretch the x-axis to about 2 million. This confirms that the mean is higher than the median.

Figure 2: Top 10 Most Common Brands (Bar Chart)



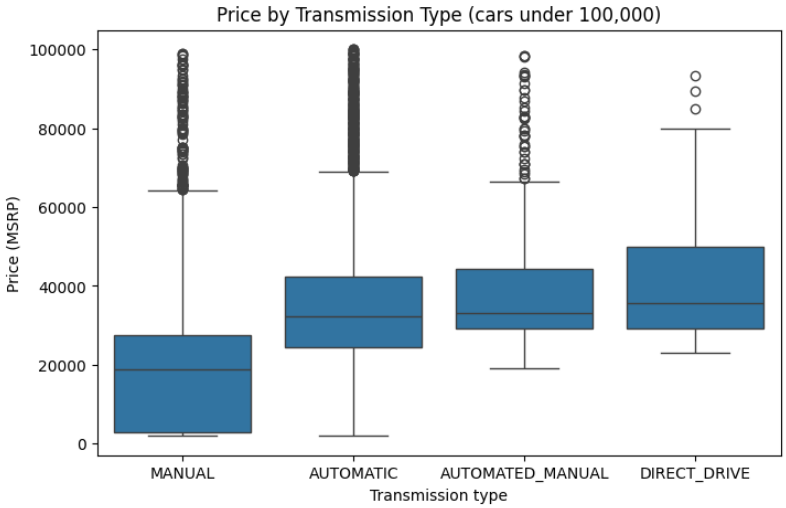
Observation: Chevrolet is the most common brand with 1,083 records, followed by Ford (825) and Toyota (719). Volkswagen, Nissan and Dodge each have more than 500. The ten brands together make up about 54% of the dataset. These counts are records in the dataset, not sales figures.

Figure 3: Cars by Transmission Type (Count Plot)



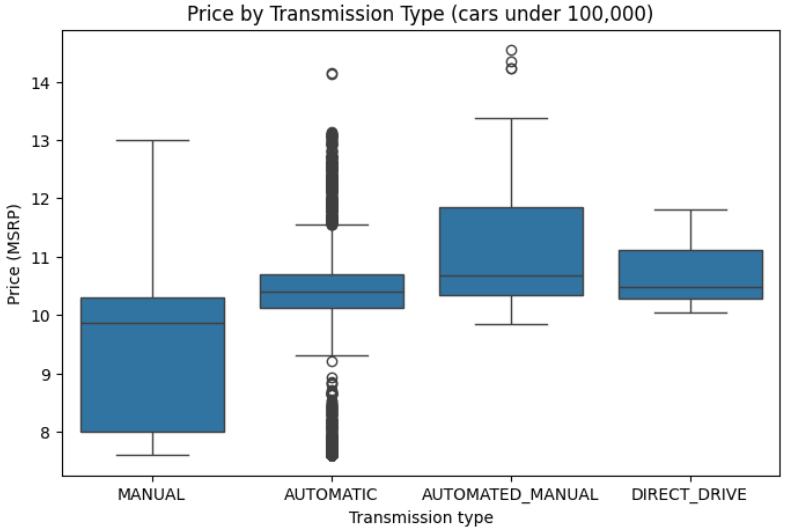
Observation: Automatic cars dominate with 7,929 records (70.9%), followed by manual with 2,634 (23.6%). Automated manual has 553 (4.9%) and direct drive only 68 (0.6%), so the last two groups are small and their statistics should be read with care.

Figure 4: Price by Transmission Type, Cars Under 100,000 (Box Plot)



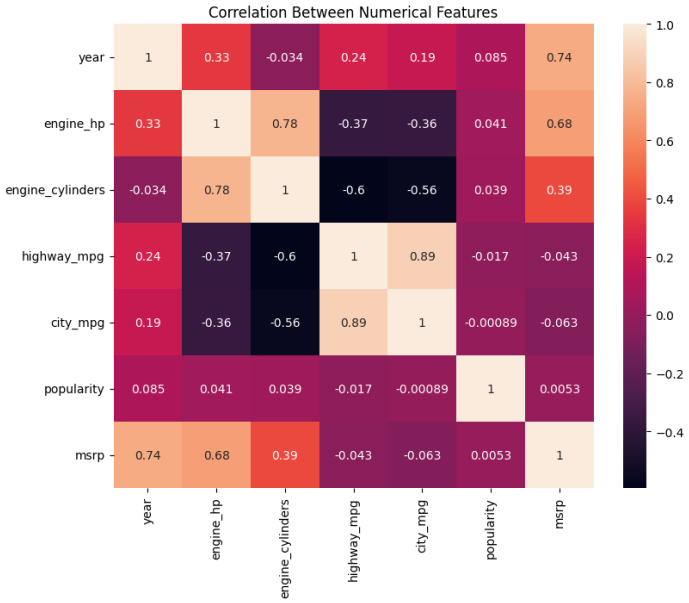
Observation: Among cars priced under 100,000, manual cars have the lowest median price (about 19,000) and the widest lower range; their lower quartile is near 3,000. Automatic cars have a median of about 32,000, and automated manual and direct drive cars have higher medians. All groups show high-price outliers. The very low manual prices are linked to a large number of records priced exactly 2,000 (see Limitations).

Figure 5: Log Price by Transmission Type (Box Plot, log scale)



Observation: On the log scale, the price differences are easier to compare. Manual cars have the lowest median (about 9.9 on the log scale, roughly 19,000) and the widest box; automatic cars are concentrated around 10.4 (roughly 33,000) with many outliers on both sides; automated manual cars have the highest median (about 10.7, roughly 44,000). Note: the chart title says "under 100,000" but, because prices were already log-transformed, the filter had no effect and all 11,184 cars are included; the y-axis shows log(MSRP).

Figure 6: Correlation Between Numerical Features (Heatmap)



Observation: Correlations with price (log MSRP): year 0.74, engine HP 0.68 and engine cylinders 0.39, so newer and more powerful cars tend to cost more. Highway and city MPG are very strongly related to each other (0.89), and engine HP and cylinders are related (0.78). Cylinders are negatively related to highway MPG (-0.60), i.e. bigger engines are less fuel-efficient. MPG and popularity have almost no relationship with price (between -0.06 and 0.01). Correlation does not prove cause.

7. Key Findings

1. **Chevrolet, Ford and Toyota are the most common brands** (1,083, 825 and 719 records). The top 10 brands account for about 54% of all records.
2. **Automatic transmission is by far the most common:** 70.9% of cars, against 23.6% manual, 4.9% automated manual and 0.6% direct drive.
3. **Prices are highly right-skewed.** The mean price (41,973) is much higher than the median (30,705), with a maximum of 2,065,902. The extreme prices belong to genuine luxury cars such as the Bugatti Veyron 16.4, Lamborghini Reventon and Maybach Landaulet.
4. **Manual cars have the lowest median price** (about 19,375) compared with automatic (33,000), direct drive (35,825) and automated manual (43,550) cars, although part of this gap comes from placeholder prices (see Limitations).
5. **Engine power and model year have the strongest link with price.** Correlation with log price is 0.68 for engine HP and 0.74 for year, and 0.39 for cylinders, while fuel efficiency and popularity show almost no relationship with price.
6. **Related features move together:** highway and city MPG (0.89), engine HP and cylinders (0.78), and cylinders and highway MPG (-0.60, more cylinders means lower efficiency).

Limitations of the dataset

- **Price is MSRP** (original list price), not resale price, so price versus year shows list-price levels and not depreciation. There is no mileage column.
- **Placeholder prices:** 743 cleaned records have a price of exactly 2,000, all from model years 2000 or earlier (450 of them manual cars). These look like default values and were not removed, so they lower the manual-car prices and increase the strength of the year-price correlation.
- **Unusual MPG values:** one record shows 354 highway MPG, and 41 records show more than 100 MPG (mostly electric cars measured in MPGe). These were kept and not treated.

- **Missing category information:** about 30% of Market Category is "Not Specified", so it was not used in the analysis.
- **Scope:** the data covers model years 1990 to 2017 only, and rows are model/trim variants, so counts are not sales volumes. Small groups (direct drive: 68 rows) limit what can be concluded.

8. Conclusion

Learning: this project showed the full EDA process: inspecting a dataset, finding and fixing missing values, duplicates and inconsistent labels, and then using charts and statistics to answer questions. The dataset is dominated by automatic, regular-unleaded cars from a few popular brands, prices are highly skewed because of a small number of luxury cars, and horsepower and model year are the features most linked with price.

Challenges: handling the very skewed price column (the log transformation made the box plot readable), choosing how to fill missing values, and noticing that some prices are default values rather than real prices.

Future analysis: remove the placeholder prices and the 354 MPG record and repeat the price comparisons; add a scatter plot of engine HP against price; compare average price by brand and by vehicle style; and, in a later project, build a model to predict price.